Lab#06

Implementation of math class with various methods in java

Objective

To get familiarity with the math class with various methods in java.

Lab Goals

To provide working insight into the usage of math class in java programming:

a. Demonstrate the implementation of math class.

b. Understand the purpose and usage of the Math class in Java.

c. Learn about the various mathematical functions and constants provided by the Math class.

Math Function

Math Class in Java contains certain methods to perform different numeric operations,for instance, *exponential, square root, logarithmic, trigonometric* functions.

Method	Description
Math.abs()	It will return the Absolute value of the given value.
Math.max()	It returns the Largest of two values.
Math.min()	It is used to return the Smallest of two values.
Math.round()	It is used to round of the decimal numbers to the nearest value.
Math.sqrt()	It is used to return the square root of a number.
Math.cbrt()	It is used to return the cube root of a number.
Math.pow()	It returns the value of first argument raised to the power to second argument.

The Java Math class has many methods that allows you to perform mathematical tasks on numbers.

Exercise

A. Write a program that generates numbers randomly between 97 to 122 and convert that number to their corresponding character (**Condition:** don't use looping).

Program:

```java
public class Lab6 {

    public static void main(String[] args) {

        // Generate random number between 97 and 122
        int num = (int)(Math.random() * 26) + 97;

        // Convert number to character
        char ch = (char) num;

        // Print result
        System.out.println("Number    : " + num);
        System.out.println("Character : " + ch);
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 6> java Lab6
Number    : 99
Character : c
PS C:\Users\Ansharah\Desktop\Java\Lab 6>
```

B. Write a program that calculates the hypotenuse of a right-angled triangle using Math.sqrt() and the Pythagorean theorem **($a^2 + b^2 = c^2$).** Take **a** and **b** as user input.

Program:

```java
import java.util.Scanner;

public class Lab6 {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Take user input
        System.out.print(s: "Enter side a: ");
        double a = sc.nextDouble();

        System.out.print(s: "Enter side b: ");
        double b = sc.nextDouble();

        // Calculate hypotenuse using Pythagorean theorem
        double c = Math.sqrt(a * a + b * b);

        // Print result
        System.out.println("Hypotenuse c = " + c);

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 6> javac Lab6.java
PS C:\Users\Ansharah\Desktop\Java\Lab 6> java Lab6
Enter side a: 56
Enter side b: 20
Hypotenuse c = 59.464274989274024
PS C:\Users\Ansharah\Desktop\Java\Lab 6>
```

C. Write a program to find the **Compound Interest** using **Math.pow():**

Formula:

$$A = P \times \left(1 + \frac{r}{n}\right)^{n \times t}$$

where:
A = final amount
P = principal amount
r = annual interest rate
n = number of times interest is compounded per year
t = time in years

Program:

```java
public class Lab6 {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter Principal (P): ");
        double P = sc.nextDouble();

        System.out.print(s: "Enter Rate (r) in %: ");
        double r = sc.nextDouble() / 100;

        System.out.print(s: "Enter Times compounded per year (n): ");
        int n = sc.nextInt();

        System.out.print(s: "Enter Time in years (t): ");
        int t = sc.nextInt();

        // Formula: A = P * (1 + r/n) ^ (n*t)
        double A = P * Math.pow(1 + r / n, n * t);
        double CI = A - P;

        System.out.println("Final Amount        : " + A);
        System.out.println("Compound Interest   : " + CI);

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 6> javac Lab6.java
PS C:\Users\Ansharah\Desktop\Java\Lab 6> java Lab6
Enter Principal (P): 1000
Enter Rate (r) in %: 5
Enter Times compounded per year (n): 12
Enter Time in years (t): 2
Final Amount        : 1104.941335558327
Compound Interest   : 104.94133555832695
PS C:\Users\Ansharah\Desktop\Java\Lab 6>
```

D. Write a program that calculates the Euclidean distance between two points (x1,y1) and (x2,y2) using **Math.sqrt()** and **Math.pow().**

Formula:

$$\text{Distance} = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$

Program:

```java
import java.util.Scanner;

public class Lab6 {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Take user input
        System.out.print(s: "Enter x1: ");
        double x1 = sc.nextDouble();

        System.out.print(s: "Enter y1: ");
        double y1 = sc.nextDouble();

        System.out.print(s: "Enter x2: ");
        double x2 = sc.nextDouble();

        System.out.print(s: "Enter y2: ");
        double y2 = sc.nextDouble();

        // Formula: distance = sqrt((x2-x1)^2 + (y2-y1)^2)
        double distance = Math.sqrt(Math.pow(x2 - x1, b: 2) + Math.pow(y2 - y1, b: 2));

        System.out.println("Distance = " + distance);

        sc.close();
    }
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 6> Javac Lab6.java
PS C:\Users\Ansharah\Desktop\Java\Lab 6> java Lab6
Enter x1: 90
Enter y1: 45
Enter x2: 36
Enter y2: 78
Distance = 63.28506932918696
PS C:\Users\Ansharah\Desktop\Java\Lab 6>
```